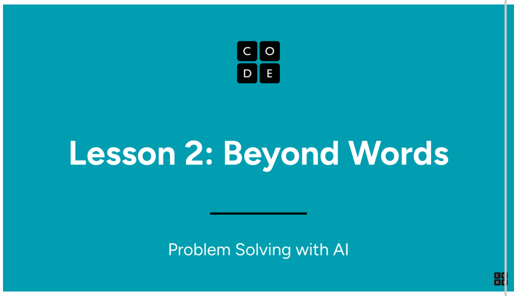
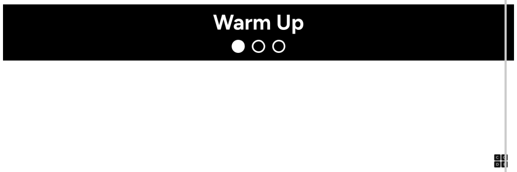
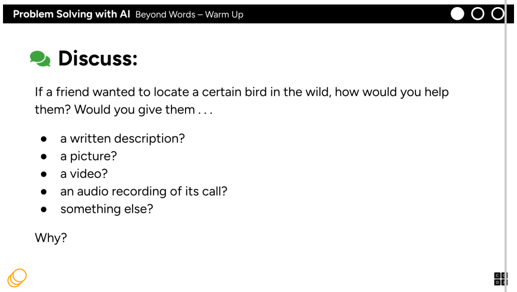
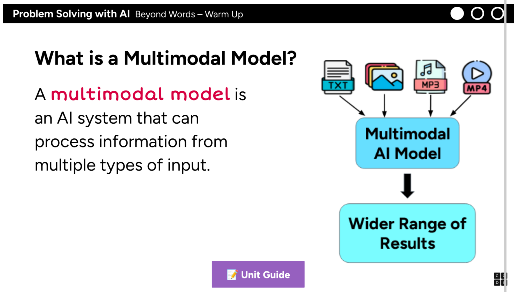
Slide	Notes
	
	Warm Up (10 mins)
	■ Discuss: Click through the animated slide to display the prompts: If a friend wanted to locate a certain bird in the wild, how would you help them? Would you give them . . . a written description? a picture? a video? an audio recording of its call? something else? Why? ■ Discussion Goal: Students should recognize that different types of media serve different purposes. Text provides details, images help with identification, and audio and video capture sound and movement. AI must process different inputs to generate useful responses. Teaching Tip If students struggle to pick a media type, prompt them with follow up questions, such as: "How do YOU recognize a bird in the wild?" or "Would text be enough to help a friend who has never seen this bird before?"
	■■ Say: AI models like GPT-4, that powers ChatGPT, are trained to analyze multiple types of input (text, images, audio). Some AI models input text only, some input text and images, some input video, etc. Our AI Chat allows us to input text, images, and PDFs. ■■ Do This: Define multimodal model. Provide students with some real-world examples, such as Google Lens, self-driving cars, ChatGPT's image recognition, etc. Teaching Tip If students ask about AI that uses all four types of input (text, image, video, and sound), acknowledge that some AI models specialize in one type while others combine them. Self-driving cars, for example, use vision and audio sensors.

Where is Multimodal AI Used?

- **Accessibility**: screen readers interpreting images and text
- **Healthcare**: AI analyzing X-ray images and text
- **Translation**: real-time text and audio speech translation
- **Security**: facial and object recognition

■■ Say: Multimodal AI isn't just used in chatbots – it's changing industries.

■■ Do This: Use the examples on the slide to show how AI uses multiple inputs in real-world applications.

Lesson Objectives

By the end of this lesson, you will be able to . . .

- Explain that multimodal AI models process information from multiple types of input.
- Experiment with different media inputs to observe AI's interpretation and limitations.
- Compare how AI-generated outputs change based on the type of input provided.
- Identify privacy and ethical concerns when using AI with multimedia data.

■■ Do This: Introduce the lesson objectives.

Question of the Day

What is the effect of using multimodal inputs with AI?

■■ Do This: Introduce the Question of the Day.

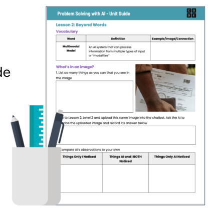
Activity

Activity (30 mins)

Unit Guide

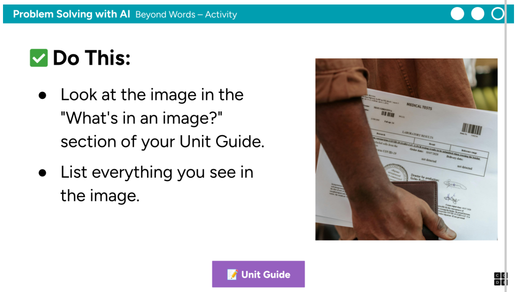
You should have:

- Problem Solving with AI Unit Guide
- pen or pencil



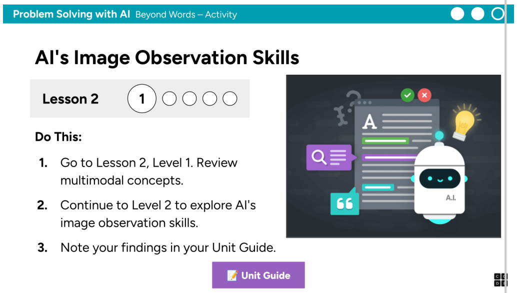
Add Images (10 mins)

■■ Do This: Have students take out their Problem Solving with AI Unit Guide.



■■ Do This: Have students list everything they observe in the image found in the "What's in an image?" section of Lesson 2 in their Unit Guides.

■ Circulate: As students are listing observations, listen for students focusing only on objects rather than details like lighting, time of day, or social context. If needed, ask: "What details beyond objects help describe this image?"

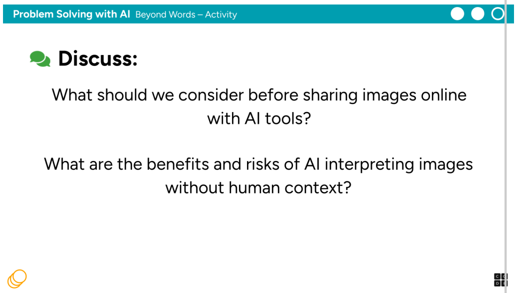


■■ Say: Now that you have noted what you see in the image, let's see how AI describes it.

■ Transition: Direct students to Lesson 2, Level 1 to complete Levels 1 and 2.
Level #1: Students review multimodal concepts
Level #2: Students add an image and ask AI Chat to describe it. Students should complete the "What's in an image?" and "Privacy and Ethical Concerns" sections in their Unit Guide.

■ Circulate: Look for students noticing unexpected AI-detected details (e.g., time of day, inferred location). If students struggle, ask students, "Did AI describe something you missed?" or "Did AI infer anything beyond what's in the image?"

Teaching Tip
If students struggle to compare their observations with AI's response, ask: "Did AI notice something you didn't? Why might that be?"



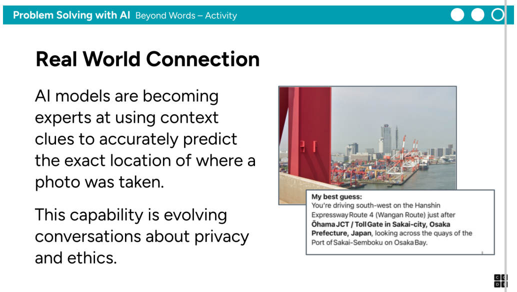
■ Discuss: Click through the animated slide to display the prompts:
What should we consider before sharing images online with AI tools?
What are the benefits and risks of AI interpreting images without human context?

■ Discussion Goal: Students should understand that AI can infer more than just visible objects in an image, including time, location, and personal details. They should critically evaluate privacy risks and ethical concerns when sharing images with AI.

Teaching Tip
For responses related to ethical considerations - If students give general answers, push for specificity by asking: "What's one real-world example of when an AI-generated image could cause harm?"

For responses related to privacy concerns - If students don't connect privacy risks to real-world consequences, ask: "How could an AI misinterpret an image and make incorrect assumptions?"

If time allows, prompt the class with: "Based on what was shared, do you think AI should be allowed to process certain types of images? Why or why not?" to get students thinking about ethics of AI tool use.



■■ Do This: Share this example of AI models increasingly having reverse image search capabilities. Providing an image of a location into newer AI models will often output a very accurate guess as to the image's location using only context clues present in the image.

Teaching Tip
This YouTube Short (https://youtube.com/shorts/qCIzSx0QGVI?si=oKNSi7LlAPgQNFlp) shows how a human uses context clues from an image to find the location within a music video. It illustrates some of the ways in which context is used to pinpoint locations. This article from Tech Crunch also details the debate about privacy and ethics that is arising from this new AI ability: https://techcrunch.com/2025/04/17/the-latest-viral-chatgpt-trend-is-doing-reverse-location-search-f

Problem Solving with AI Beyond Words – Activity

Partner Prompt Practice

Partner A: Prompt Designer

- Write the prompt for AI Chat.
- Be clear about your goal (what you want the AI to do)
- After testing, listen to feedback and improve your prompt.

Partner B: Tester and Explainer

- Enter Partner A's prompt into AI Chat.
- Explain what worked and what didn't.
- Work with Partner A to suggest changes for the next prompt.

■■ Do This: Pair up students and assign them roles. One partner writes prompts (Prompt Designer). The other partner tests and critiques results (Tester & Explainer).

■■ Say: You will work in pairs for this next activity. Here is an example of how these two roles might work together:

Step 1: Partner A's Prompt

"Write a poem about space."

Step 2: Partner B Tests & Explains

Runs the prompt.
Explains: "The AI gave a poem, but it's very short and doesn't use much imagery. It also doesn't rhyme, if we want that."

Step 3: Partner A Revises

"Write a rhyming poem about space with vivid imagery, at least 8 lines long."

Teaching Tip
You may choose to replace the example above with your own or one that is more closely aligned with your students experience and interests. If you have time, you may also choose to ask for a student volunteer and then model this partner work in front of the class one time before having students move on to the next level.

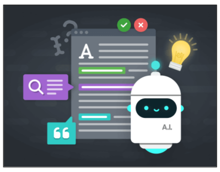
Problem Solving with AI Beyond Words – Activity

Scenario-Based Prompting

Lesson 2 3

Do This:

1. Go to Lesson 2, Level 3.
2. In partners, explore how AI-generated responses change based on the input type.



Scenario-Based Multimodal Prompting (20 mins)

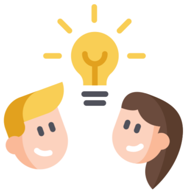
■ Transition: Direct students to Lesson 2, Level 3. Students work with a partner to prompt the chatbot to suggest improvements, first based off of a text-only description and then based on an image.

■ Circulate: Watch for students who struggle to explain why AI's response changed when using text vs. images. If needed, ask: "What new details did AI include when you used an image? Was the response more specific or more general?"

Problem Solving with AI Beyond Words – Activity

Check Your Understanding

Lesson 2 ◯◯◯ 4 ◯



Do This:

1. Go to Lesson 2, Level 4.
2. Respond to the question.

■■ Do This: Direct students to Lesson 2, Level 4 to complete the formative assessment.

■■ Do This: Display students' anonymous responses, and choose one of the three approaches below based on how students did:
Option 1: Whole-Class Poll & Debrief (Quick Check-In)
Display the poll results for the multiple choice question.
Call on 2-3 students from each answer group (Text Only vs Text +Image) to explain their reasoning
Wrap up by asking students: “How does this confirm or challenge what we learned about effective prompting?”
Option 2: Small Group Discussion & Class Share-Out (Collaborative)
Assign students to groups of 3-4.
Each group discusses: How did AI’s response change when you included an image? and What do you think influenced AI’s different responses?
Groups write down 1-2 insights on a shared board (physical or digital).
Teacher facilitates a brief whole-class discussion based on key themes.
Option 3: Silent Gallery Walk (Reflective)
Post anonymized student responses using the “View Student Responses” Pin Response feature.
Students independently respond to pinned responses using:
Agree/Disagree: Do they agree with the reasoning? Why?
Question: What would they ask this student to clarify?
Connection: How does this relate to prior learning?
End with whole-class synthesis: What patterns or surprises did students notice?

Assessment Opportunity
This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:
If most students correctly identified how AI's response changed with an image, reinforce key takeaways with a brief discussion and move forward to the choice-level scenario activity.
If responses indicate confusion about why AI responded differently to text vs. image inputs, ask students to revisit AI’s descriptions and reflect on what details were added or changed.
If students struggle to connect this to effective prompting, review the warm up discussion about media choices and ask: "How does this activity confirm what we said about text vs. image information?"

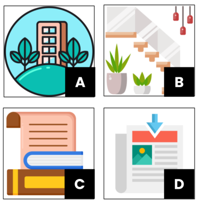
Problem Solving with AI Beyond Words – Activity

Practice with a Multimodal Model

Lesson 2 ◯◯◯◯ 5

Do This:

1. Go to Lesson 2, Level 5.
2. Choose a scenario and practice prompting the AI for suggestions using images, text, and PDFs.



■■ Say: Now that we've discussed your findings, let's practice using multimodal AI in a scenario!

■ Transition: Direct students to Lesson 2, Levels 5 to pick one of the scenarios to practice prompting the AI for suggestions using images and text.

■ Circulate: Ensure students recognize how AI's response shifts when using text vs. images. If students only describe objects AI listed, ask: "Did AI provide any deeper insights beyond what you expected?"

Teaching Tip
Encourage students to experiment with different combinations of text, images, and PDFs to see how AI's responses change. If students finish early or need an extra challenge, suggest trying Option D: Investigative Journalism, where they can analyze how AI generates headlines and summaries from different sources. Have them compare AI-generated headlines with real-world news articles on the same topic. For an added challenge, they can test how changing a single keyword, altering the image, or adding a conflicting source affects AI's response and credibility.

Problem Solving with AI Beyond Words – Activity **Discuss:** What key details did AI focus on in the image? When you prompted AI Chat with an image ***and*** a PDF, how did its response change?	■ Discuss: What key details did AI focus on in the image? When you prompted AI Chat with an image and a PDF, how did its response change? ■ Discussion Goal: Students should recognize that AI was able to pull out key details from the images. Adding in a PDF helped give additional context to AI's response and make it more accurate. Teaching Tip Not all students interacted with an AI Chat level that required them to add a document. As you circulate during students' work time, find students who added both an image and a PDF to call on during the discussion.
Wrap Up	Wrap Up (5 mins)
Problem Solving with AI Beyond Words – Wrap Up **Today, you learned about . . .** • How multimodal AI models process information from multiple types of input. • How different media inputs affect AI's interpretation and limitations. • How AI-generated outputs change based on the type of input provided. • Privacy and ethical concerns when using AI with multimedia data.	■■ Do This: Review the lesson objectives.
Problem Solving with AI Beyond Words – Wrap Up **Question of the Day** What is the effect of using multimodal inputs with AI? Unit Guide	■■ Do This: Direct students to their Unit Guides to answer the Question of the Day. Give them 30 seconds to write a response using one of the sentence starters found in the "Hint' of the Question of the Day for this lesson before having students share with a partner or ask several students to share with the class. Teaching Tip This works well as an exit ticket or a quick way to gauge comprehension during a lesson. If time allows, guide the class discussion toward real-world applications. Use these prompts if needed: Where have you seen multimodal AI used outside of chatbots? How might multimodal AI impact careers in accessibility, medicine, or security? What are examples of AI tools that use images and text together? (e.g., Google Lens, AI-generated captions, AI medical imaging analysis) Assessment Opportunity Use student responses to gauge understanding. If students struggle to explain how AI's response changed based on input type, revisit earlier examples and prompt students to compare outputs with and without images. If most responses show a strong grasp, move forward.

Key Vocabulary

- **multimodal model:** an AI system that can process information from multiple types of input

■■ Do This: Review the Key Vocabulary.